

[Aim: 100/100 in Maths]

आरंभ

QUADRATIC EQUATIONS

Lecture 8

← last lecture

$$\begin{array}{c} \uparrow \quad \uparrow \\ \text{26} = 2 \times \underline{10} + 6 \times \underline{1} \\ \text{T} \quad \text{U} \end{array}$$

$$\begin{array}{c} \uparrow \quad \uparrow \\ \text{73} = 7 \times 10 + 3 \times 1 \\ \text{T} \quad \text{U} \end{array}$$

$$\begin{array}{l} U = y \\ T = x \end{array} \rightarrow N = x \times 10 + y \times 1$$

$$\boxed{N = 10x + y}$$

digits
reverse

$$\begin{array}{l} U = x \\ T = y \end{array} \rightarrow N' = y \times 10 + x \times 1$$

$$\boxed{N' = 10y + x}$$

#LP: A 2-digit number is such that the product of its digits is 24. If 18 is subtracted from the number, the digits interchange their places. Find the number.

Let unit place digit = y
tens place digit = x

$$\text{No} \Rightarrow 10x + y$$

$$N = 10x + y$$

Reverse
digits

$$U = x$$

$$T = y$$

$$N' = 10y + x$$

ATQ

$$xy = 24 \quad \text{--- (I)}$$

ATQ

$$N - 18 = N'$$

$$10x + y - 18 = 10y + x$$

$$\Rightarrow 10x - x + y - 10y = +18$$

$$\Rightarrow 9x - 9y = 18$$

$$\Rightarrow 9(x - y) = 18$$

$$\Rightarrow x - y = 2 \quad \text{--- (II)}$$

$$xy = 24$$

$$x = \frac{24}{y}$$

$$x - y = 2$$

$$\frac{24}{y} - y = 2$$

$$\frac{24 - y^2}{y} = 2$$

$$24 - y^2 = 2y$$

$$y^2 + 2y - 24 = 0$$

$$y = 4$$

$$x = 6$$

Final Ans
 $x = 10, x + y.$

WORK ATM Sawaal:-

एक टैंक

#LP: If two pipes function simultaneously, a reservoir will be filled in 12 hours.

One pipe fills the reservoir 10 hours faster than the other. How many hours will the second pipe take to fill the reservoir?

Two pipes together, time = 12 hr

Separately

let time taken by small tap to fill reservoir completely = x hr

x hr $\rightarrow$ 1 fill

1 hr $\rightarrow \frac{1}{x}$ fill

12 hr $\rightarrow \left(12 \times \frac{1}{x}\right)$ fill

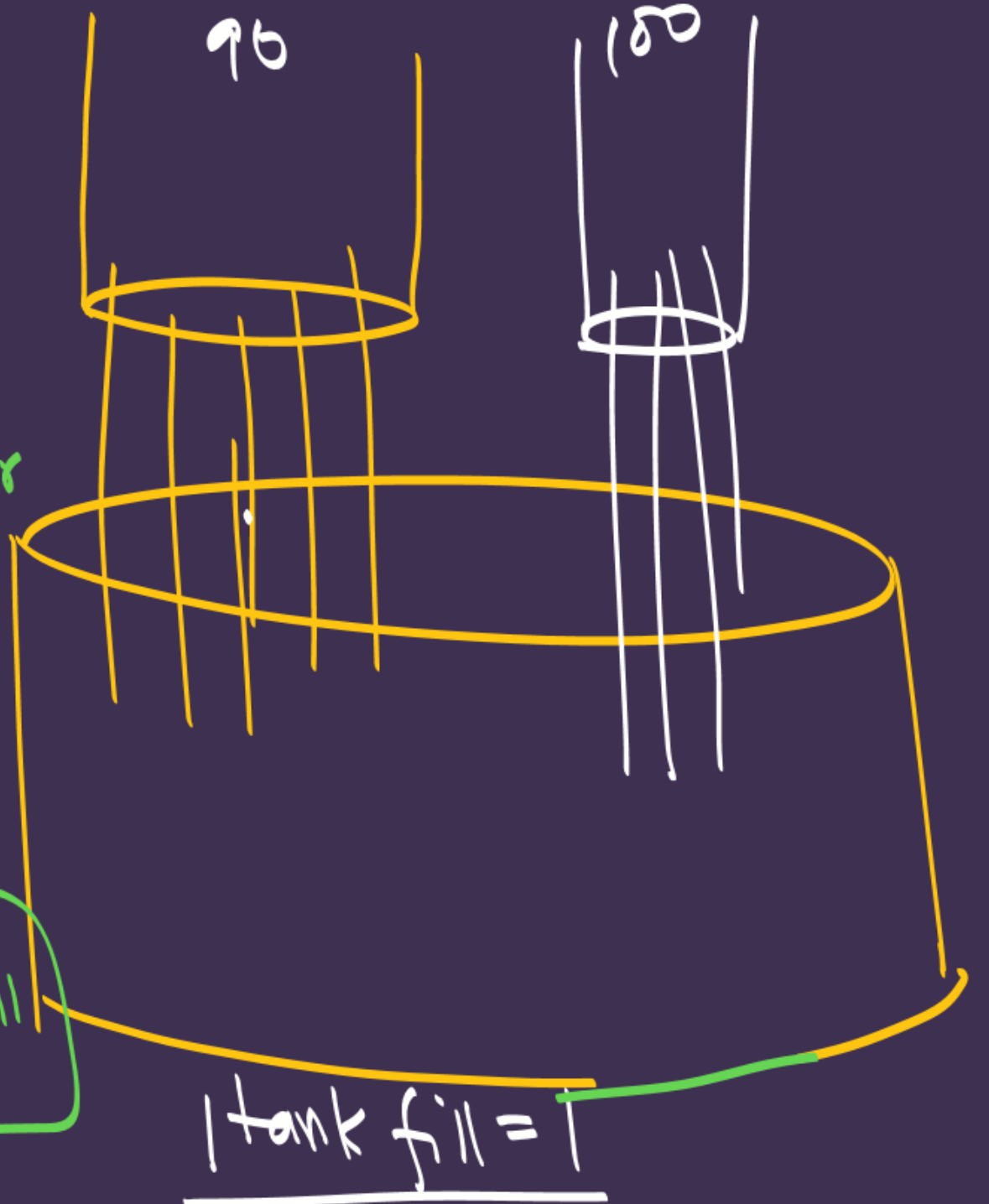
Separately

time taken by big tap to fill tank completely = $(x-10)$ hr

$(x-10)$ hr $\rightarrow$ 1 fill

1 hr $\rightarrow \frac{1}{x-10}$

12 hr $\rightarrow \left(12 \times \frac{1}{x-10}\right)$ fill



for 12hrs simultaneously

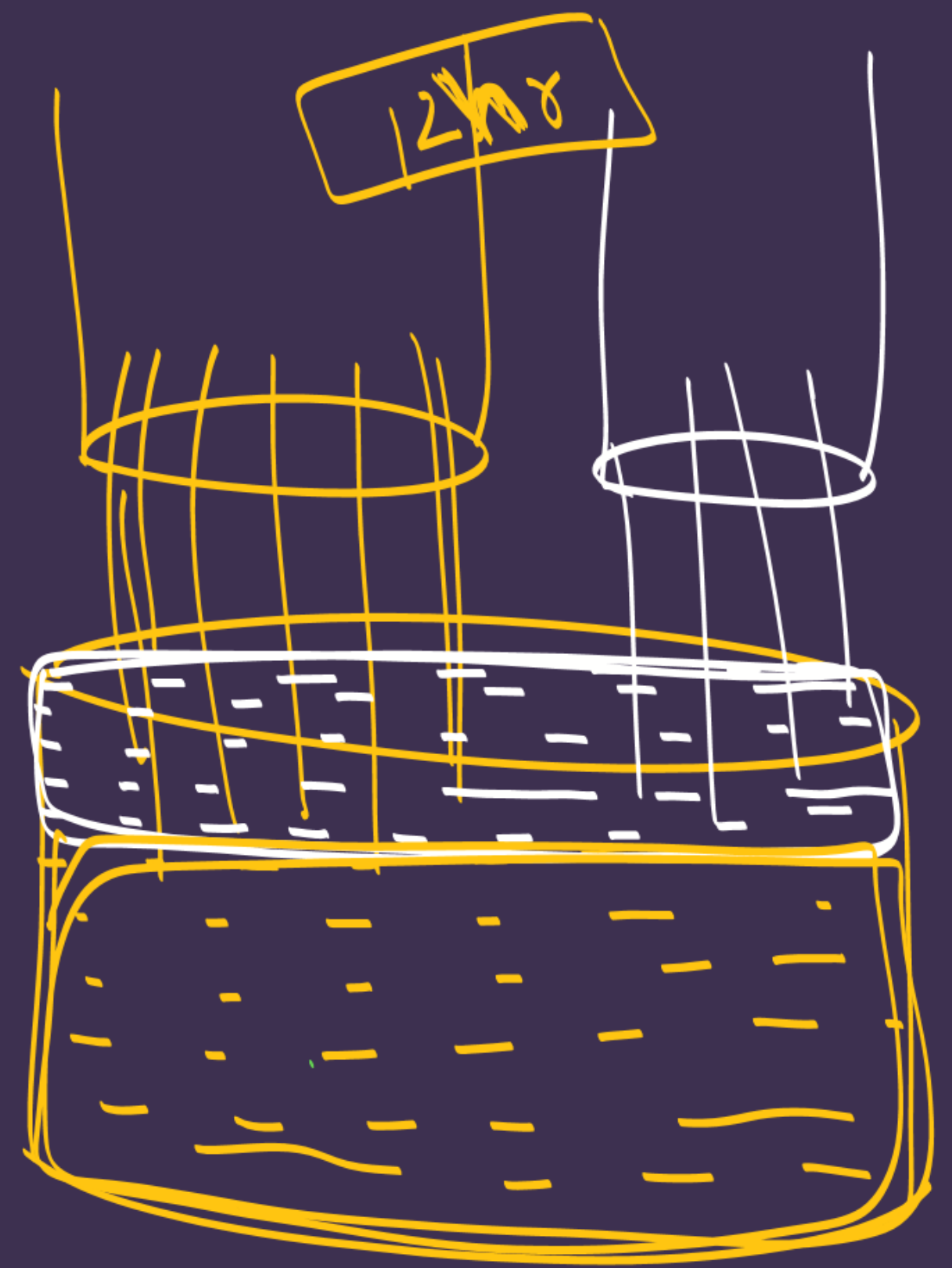
$$(W.D)_{big} + (W.D)_{small} = 1$$

$$\Rightarrow \left(\frac{12}{x-10}\right) + \left(\frac{12}{x}\right) = 1$$

$$\Rightarrow 12 \left(\frac{1}{x-10} + \frac{1}{x} \right) = 1$$

$$\Rightarrow \frac{1(x) + 1(x-10)}{x(x-10)} = \frac{1}{12}$$

$$\Rightarrow \frac{x + x - 10}{x^2 - 10x} = \frac{1}{12} \quad \text{solve}$$



Total W.D = ①

#LP: A takes 10 days less than the time taken by B to finish a piece of work.
If both A and B together can finish the work in 12 days, find the time taken by B to finish the work.

Individually

(A)

(B)

(A) 10 day less than B

∴ time taken by (A) to complete work separately = $(x-10)$ days

$(x-10)$ days $\rightarrow$ 1 work

1 day $\rightarrow \frac{1}{x-10}$ work

12 days $\rightarrow 12 \times \frac{1}{x-10}$ work

(B) let time taken by (B) to completely work separately = x days

x days $\rightarrow$ 1 work

1 day $\rightarrow \frac{1}{x}$ work

12 day $\Rightarrow 12 \times \frac{1}{x}$ work

Together time = 12 days

For 12 days

total W.D = 1

$(W.D)_A + (W.D)_B = 1$

$\left(\frac{12}{x-10}\right) + \left(\frac{12}{x}\right) = 1$

↓ solve x ✓

#LP: Statement-1: If $5+\sqrt{7}$ is a root of a quadratic equation with rational coefficients, then its other root is $5-\sqrt{7}$. (T)

Statement-2: surd roots of a quadratic equation with rational coefficients occur in conjugate pairs. (T)

$a, b, c \rightarrow \mathbb{Q}$

$$ax^2 + bx + c = 0$$

$$x + y\sqrt{z}$$

$$x - y\sqrt{z}$$

Quadratic
Equation

शुक्रिया !!

THANK YOU COODIESS !!

HW → complete module

